MATHEMATICS PROJECT WORK RECORD :: FA-2

Topic: Statistics

1. PRELIMINARY INFORMATION

Name of the Student:	.....
Class & Section:	.....
Roll Number:	.....
Subject:	**Mathematics**
Topic:	**Statistics**

2. AIM

To record the test marks scored by 30 classmates, organize the data into a grouped frequency distribution, calculate the Mode using the standard statistical formula, and verify the modal value visually using a histogram.

3. OBJECTIVES

- Collect primary test score data from 30 students.
- Group the raw marks into continuous class intervals of uniform width (h = 10).
- Identify the modal class containing the highest frequency (f_1).
- Calculate Mode using: Mode = $l + [(f_1 - f_0) / (2f_1 - f_0 - f_2)] \times h$.
- Verify the Mode graphically through the intersection of diagonal lines in the modal rectangle of a histogram.

4. MATERIALS REQUIRED

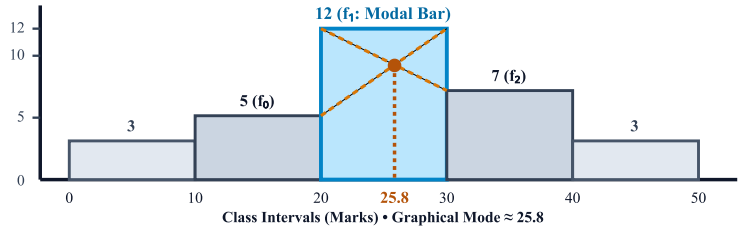
Project recording paper, graph sheet, 30 cm ruler, pencil, eraser, pen, and Class 10 Mathematics textbook.

5. PROCEDURE

1. Collected test marks (out of 50) of 30 classmates and tabulated them.
2. Formed continuous intervals: 0–10, 10–20, 20–30, 30–40, and 40–50.
3. Identified the modal class with the highest frequency: interval 20–30 (frequency = 12).
4. Extracted values: lower boundary $l = 20$, $f_1 = 12$, preceding frequency $f_0 = 5$, succeeding frequency $f_2 = 7$, and width $h = 10$.
5. Substituted values into the formula to find the exact numerical Mode.
6. Constructed a clean histogram and plotted diagonals across adjacent bar tops to confirm the Mode graphically.

6. OBSERVATIONS AND TABULATION

Class Interval (Marks)	Frequency (f)	Designation	Remarks
0 – 10	3	—	Initial Class
10 – 20	5	f_0	Frequency of Preceding Class
20 – 30	**12**	**f_1**	**Modal Class (Highest Frequency)**
30 – 40	7	f_2	Frequency of Succeeding Class
40 – 50	3	—	Terminal Class
Total	**N = 30**	**—**	**Total Sample Size**



Step-by-Step Calculation:

• Modal class = 20–30 | $l = 20$, $f_1 = 12$, $f_0 = 5$, $f_2 = 7$, $h = 10$

• Mode = $l + [(f_1 - f_0) / (2f_1 - f_0 - f_2)] \times h$

• Mode = $20 + [(12 - 5) / (2(12) - 5 - 7)] \times 10 = 20 + [7 / (24 - 12)] \times 10$

• Mode = $20 + [7 / 12] \times 10 = 20 + 5.83 =$ **25.83**

7. CONCLUSION

- The calculated Mode of the marks distribution is **25.83**.
- The graphical determination of mode using diagonals in the histogram accurately confirms the calculated value.

8. ACKNOWLEDGEMENT

I express my sincere thanks to my **Mathematics Teacher** for continuous guidance and my classmates for providing their test marks.

9. RESOURCES

Telangana SCERT Class 10 Mathematics Textbook (Chapter: *Statistics*) and classroom score charts.

Signature of Student
Date: ____/____/2026

Signature of Mathematics Teacher
Verified